

LESSON 13.4

SOLVING FOR UNKNOWN ANGLES IN TRIANGLES



LESSON INTRODUCTION

MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.

LEARNING TARGETS

- I can write and solve algebraic equations to solve mathematical problems involving angle relationships of triangles.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships and theorems of triangles.

ESSENTIAL QUESTIONS

- How can I use angle relationships of triangles to write and solve algebraic equations for problems involving angles of triangles?

LESSON NARRATIVE

In this lesson, students extend their learning from the prior lesson to solve for unknown angles in triangles. Students collaborate with their peers to find missing angle measures using what they've learned about angle relationships with vertical, complementary, supplementary, and adjacent angles. Students will be introduced to writing equations that can be used to find the value of an unknown angle of a triangle. Students will use multiple-angle relationships to find missing angles.

COMPONENT SUMMARY

13.4.1 WARM-UP (~5 MIN)

Respond to a video of a museum exhibit

13.4.3 GUIDED INSTRUCTION (10 MIN)

Write equations to find missing angles in triangles

13.4.5 YOUR TURN! (10 MIN)

Write and solve equations to find angle measures of triangles

13.4.2 COLLABORATION (10 MIN)

Match angle relationships and descriptions to images of angles

13.4.4 COLLABORATION (10 MIN)

Use multiple-angle relationships to find missing angles

13.4.6 WRAP-UP (~5 MIN)

Write and solve an equation to find an interior angle measure

of a triangle

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Markers

ADDITIONAL PREPARATION

- 13.4.1 Warm-Up contains a video to accompany the question prompts

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize the angle relationship that exists, or may not correctly reflect the relationship in the equations they write.
- Students may believe they are solving for an angle measure when solving for a variable. Reinforce for students to pay attention to what the question is asking and what the variable represents.

THINGS TO CONSIDER

- 13.4.2 Collaboration is a refresher of the content addressed in the first lesson of this unit.
- Students have done some of the work in 13.4.3 Guided Instruction in Lesson 3.
- In 13.4.4 Collaboration and beyond, students apply what they have learned throughout this unit so far to solve problems that involve multiple angle relationships.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

13.4.2 Collaboration provides an opportunity for a Think-Pair-Share protocol to be leveraged. Students can work independently to complete the table and then pair up with a partner to review their answers and discuss to come to consensus on a final answer.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

13.4.4 Collaboration and beyond has students apply what they've learned throughout this unit to solve problems that involve multiple angle relationships. In these activities, students will decompose these complex problems into manageable parts by creating plans and procedures to logically order events, steps, or ideas.

DIFFERENTIATE INSTRUCTION

The content of this lesson requires a solid understanding of the content from prior lessons in this unit. Consider using the time during 13.4.2 Collaboration to work with students in small groups who may need additional support with angle relationships. This can be replicated during 13.4.3 Guided Instruction to work with students in small groups who may need additional support with writing and solving equations. During 13.4.4 Collaboration and beyond, use student observation to determine which students may need modified problems to scaffold their experience to the demands of the problems. Consider using markers to highlight angles that students are working with so that the angle relationships are easier to see.

13.4.1 WARM-UP: RIDING ON SQUARE WHEELS

TEACHER MOVES

To help students better visualize what they are reading about in this activity, consider playing the video before releasing students to answer the questions in the activity. This way students will actually be able to see how riding the tricycle looks.

TEACHER MOVES

Question 1 is a purposefully open-ended question. It's not the expectation that students can explain the mathematics behind this, but they can make some interesting observations nonetheless.



13.4.1 Warm-Up: Riding on Square Wheels

The National Museum of Mathematics has an exhibition called the Square-Wheel Trike. The exhibition is pictured below with two children riding on the square-wheeled tricycles. The exhibit's description says, "Take a smooth ride on square wheels."



- Describe any angles or shapes observed in the exhibition.
Answers will vary. Right angles, squares, partial cylinders
- Why do you think it's possible for the ride to be smooth on square wheels?
Answers will vary. The bicycle is ridden on a track with ridges (not smooth). The material of the wheels is like the materials on the track.



13.4.2 Collaboration: Find that Angle

1. Use the words and angle measures from the bank to complete the table below.

adjacent	common	complementary	90°
equal	supplementary	vertical	180°

Image	Angle Relationship	Description
	<u>Vertical</u> angles	∠1 and ∠2 have <u>equal</u> measures.
	<u>Complementary</u> angles	∠1 and ∠2 sum to <u>90°</u> .
	<u>Supplementary</u> angles	∠1 and ∠2 sum to <u>180°</u> .
	<u>Adjacent</u> angles	∠1 and ∠2 have a <u>common</u> vertex and side.

13.4.2 COLLABORATION: FIND THAT ANGLE

TEACHER MOVES

This activity provides an opportunity for a Think-Pair-Share protocol to be leveraged. Students can work independently to complete the table, and then pair up with a partner to review their answers and discuss to come to consensus on a final answer.

ENRICHMENT

- Why wouldn't these angles be something other than adjacent?
- What makes it different than the supplementary and complementary angle images?

13.4.3 GUIDED INSTRUCTION: WRITING EQUATIONS TO FIND MISSING ANGLES IN TRIANGLES

ENRICHMENT

- What do we know about the measures of the interior angles of a triangle?
- How can that information be helpful to solving this problem?

TEACHER MOVES

- Students may struggle with several aspects of question 1. Students may not realize that the angle marking for $\angle FMS$ is 90° .
- Students may make calculation errors during the solving process.
- Students may forget to include the $1c$ for $\angle FMS$ in their setup of the equation.



13.4.3 Guided Instruction: Writing Equations to Find Missing Angles in Triangles

Algebraic equations can be written to express the relationship between the interior and exterior angles of triangles. The equations can then be solved to find the measures of unknown angles.

1. Triangle SMF is shown.



- a. Complete the statement.

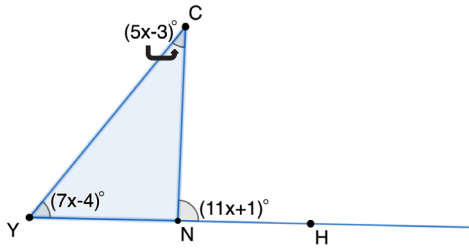
The relationship of the ☐ exterior ☒ interior angles of the triangle can be used to find the measures of the missing angles.

- b. Write an equation that can be used to find the value of c .
 $c + 90 + (6c + 6) = 180$
- c. Solve the equation.
 $c + 90 + 6c + 6 = 180$
 $c + 6c + 90 + 6 = 180$
 $7c + 96 = 180$
 $7c = 84$
 $c = 12$

- d. Use the solution to find the measure of each angle.

Angle	Measure
$\angle FSM$	12°
$\angle SFM$	$6(12) + 6 = 78^\circ$

2. Triangle CYN is shown.



- a. Complete the statement.

The relationship between the ☐ exterior
☐ interior angles of the triangle can be used to find the measures of the missing angles.

- b. Write an equation that can be used to find the value of x .
 $(11x + 1) = (7x - 4) + (5x - 3)$

- c. Solve the equation.

$$\begin{aligned} 11x + 1 &= 7x - 4 + 5x - 3 \\ 11x + 1 &= 7x + 5x - 4 - 3 \\ 11x + 1 &= 12x - 7 \\ 11x &= 12x - 8 \\ -1x &= -8 \\ &= 8 \end{aligned}$$

- d. Use the solution to find the measure of each angle.

Angle	Measure
$\angle NCY$	$5(8) - 3 = 37^\circ$
$\angle CYN$	$7(8) - 4 = 52^\circ$
$\angle CNH$	$11(8) + 1 = 89^\circ$

13.4.3 GUIDED INSTRUCTION: WRITING EQUATIONS TO FIND MISSING ANGLES IN TRIANGLES (CONTINUED)

ENRICHMENT

- What do we know about the measures of two of the interior angles of a triangle in relation to the exterior angle?
- How can that information be helpful to solving this problem?

ENRICHMENT

- How can you confirm that your answers are correct?
- What must be true when the values are substituted?

QUESTIONING

For an additional challenge, have students determine the interior angle measure of $\angle CNY$.

13.4.4 COLLABORATION: USING MULTIPLE-ANGLE RELATIONSHIPS TO FIND MISSING ANGLES

TEACHER MOVES

This activity provides students an opportunity to begin applying what they know about interior and exterior angles of triangles to solve more complex problems.

DIFFERENTIATE INSTRUCTION

Consider having students use different colored markers to highlight the angles to help students visualize the angle relationships.

TEACHER MOVES

Allow students to share out their answers to question 1b. It can benefit students to hear from their peers on their approaches to this problem and their strategy. Elicit student responses from multiple students and ask them to justify their responses.

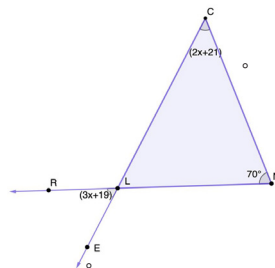
DIFFERENTIATE INSTRUCTION

For students who need help getting started with question 2, consider asking them to determine the angle relationships they see in the triangle.



13.4.4 Collaboration: Using Multiple-Angle Relationships to Find Missing Angles

- Use the figure shown to complete the following with your partner.



- Complete the table by writing the angle relationship for each set of angles using one of the phrases from the bank below. Then, complete the description of the relationship between the angles.

exterior angles to $\angle CLN$ supplementary angles
interior angles of triangle CLN vertical angles

Angles	Angle Relationship	Description of Relationship
$\angle RLE$ and $\angle CLN$	<i>Vertical angles</i>	Angles measures are <u>equal</u> .
$\angle RLE$ and $\angle RLC$	<i>Supplementary angles</i>	Angles sum to <u>180°</u> .
$\angle RLC$ or $\angle ELN$	<i>exterior angles to $\angle CLN$</i>	<u>Equal</u> to the sum of $\angle NCL$ and $\angle LNC$.
$\angle CLN$, $\angle LNC$, and $\angle NCL$	<i>interior angles of triangle CLN</i>	Angles sum to <u>180°</u> .

- Different combinations of the relationships described in the table above can be used to find the value of x . Discuss with your partner which relationships you prefer to use to find the value of x .

Answers will vary. One interior angle is given so it may be easier to add all three interior angles. Using the two supplementary angles involves 2 expressions instead of 3.

- Write and solve an equation to find the value of x .

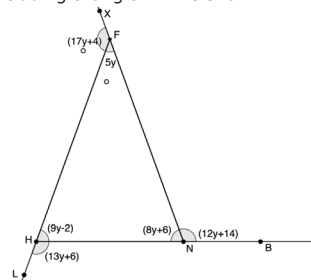
$$\begin{aligned} 3x + 19 + 2x + 21 + 70 &= 180 \\ 3x + 2x + 19 + 21 + 70 &= 180 \\ 5x + 110 &= 180 \\ 5x &= 70 \\ x &= 14 \end{aligned}$$

- Find the measure of $\angle CLN$.

$$\begin{aligned} \angle CLN \text{ is equal to } \angle RLE \text{ so } \angle CLN &= 3x + 19 \\ 3x + 19 &= 3(14) + 19 = 61^\circ \end{aligned}$$

$$m\angle CLN = 61^\circ$$

- A figure including triangle FHN is shown.



- a. The three given equations can be used to find the value of y using the angle relationships in the figure. Identify which angle relationship was used to write each equation.

Equation	Angle Relationship
$13y + 6 + 9y - 2 = 180$	◦ exterior angle of a triangle ◦ interior angles of a triangle ◦ supplementary angles
$12y + 14 = 9y - 2 + 5y$	◦ exterior angle of a triangle ◦ interior angles of a triangle ◦ supplementary angles
$5y + 9y - 2 + 8y + 6 = 180$	◦ exterior angle of a triangle ◦ interior angles of a triangle ◦ supplementary angles

- b. Choose one of the equations from Part A and use it to find the value of y .
- $12y + 14 = 9y - 2 + 5y$
 $12y + 14 = 14y - 2$
 $12y = 14y - 16$
 $-2y = -16$
 $y = 8$

- c. Find the value of each angle measure in the table.

Angle	Measure
$\angle HFX$	$17y + 4 = 17(8) + 4 = 140^\circ$
$\angle HNF$	$8y + 6 = 8(8) + 6 = 70^\circ$
$\angle NHL$	$13y + 6 = 13(8) + 6 = 110^\circ$

13.4.4 COLLABORATION: USING MULTIPLE-ANGLE RELATIONSHIPS TO FIND MISSING ANGLES (CONTINUED)

ENRICHMENT

How do you know your identified angle relationship is correct?

ENRICHMENT

- How can you confirm your answer is correct for question 2c?
- What other angles relationships could you use to verify the correct measurements were found?

13.4.5 YOUR TURN!

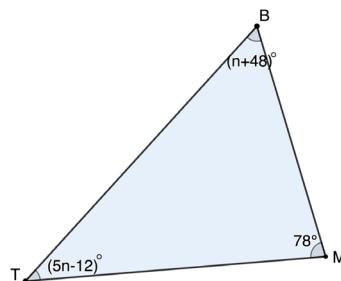
DIFFERENTIATE INSTRUCTION

Consider using this time to work with students who struggled in 13.4.4 Collaboration. This question requires less breakdown than in the prior activity, so it can provide an opportunity to deepen their understanding of this concept with interior angles without having the use of the exterior angle information. Consider modifying this question with a small group to include the measure of any of the exterior angles.



13.4.5 Your Turn!

1. Triangle BTM is shown.



- a. Write and solve an equation to find the value of n .

$$\begin{aligned}(n + 48) + (5n - 12) + 78 &= 180 \\ n + 48 + 5n - 12 + 78 &= 180 \\ n + 5n + 48 - 12 + 78 &= 180 \\ 6n + 114 &= 180 \\ 6n &= 66 \\ n &= 11\end{aligned}$$

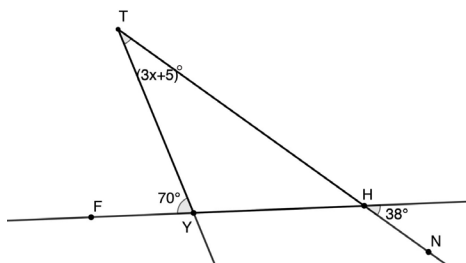
- b. Find the measure of angle BTM .

$$\begin{aligned}5n - 12 &= \\ 5(11) - 12 &= 43^\circ \\ m\angle BTM &= \underline{43^\circ}\end{aligned}$$



13.4.6 Wrap-Up: Ticket Out the Door

1. Use the figure shown to complete the following.



- a. Write and solve an equation to find the value of x .

$$\angle YHT = 38 \text{ (vertical angles are equal)}$$

$$\angle HYT = 110 \text{ (supplementary angles sum 180)}$$

$$3x + 5 + 110 + 38 = 180$$

$$3x + 153 = 180$$

$$3x = 27$$

$$x = 9$$

- b. Find the measure of $\angle HTY$.

$$3x + 5 = 3(9) + 5 = 32^\circ$$

$$m\angle HTY = \underline{32^\circ}$$

13.4.6 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

A common error that students might make is that they will assume that x is $m\angle HTY$ instead of $3x+5$. Have those students take the sum of x and the other two angles of $\triangle HTY$ to show that the result would be $9 + 110 + 38 = 157$, and not 180° .